Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 27	
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Batch	2025-29
Class	CSE A2
Academic Year & Semester	2026-27, III
Date of Submission	27/10/26
Title of Assignment:	Develop a Java application demonstrating socket programming for communication between a client and a server.

Practice questions
 	
Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p
	1.Create a simple client-server chat application using sockets.
	2. Create a Java program using socket programming where the client sends a message to the server, and the server returns the same message back (echo system).
	Server.java
	
	package *S*erver_*C*lient;
	import java.io.*;
	import java.net.*;
	
	public class Server {
	public static void main(String[] args) {
	try {

```
            ServerSocket serverSocket = new ServerSocket(5000);
            System.out.println("Server is waiting for client...");

            Socket socket = serverSocket.accept();
            System.out.println("Client connected.");

            BufferedReader in = new BufferedReader(
               new InputStreamReader(socket.getInputStream()));

            PrintWriter out = new PrintWriter(
               socket.getOutputStream(), true);

            String message;

            while ((message = in.readLine()) != null) {
               if (message.equalsIgnoreCase("exit")) {
                  break;
               }

               System.out.println("Client: " + message);
               out.println("Echo from server: " + message);
            }

            socket.close();
            serverSocket.close();
            System.out.println("Server closed.");

         } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
         }
      }
   }
```

Client.java

```
package Server_Client;
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class Client {
   public static void main(String[] args) {
      try {
         Socket socket = new Socket("localhost", 5000);
         System.out.println("Connected to server.");

         BufferedReader in = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

         PrintWriter out = new PrintWriter(
            socket.getOutputStream(), true);

         Scanner sc = new Scanner(System.in);

         while (true) {
            System.out.print("Enter message (exit to quit): ");
            String message = sc.nextLine();

            out.println(message);

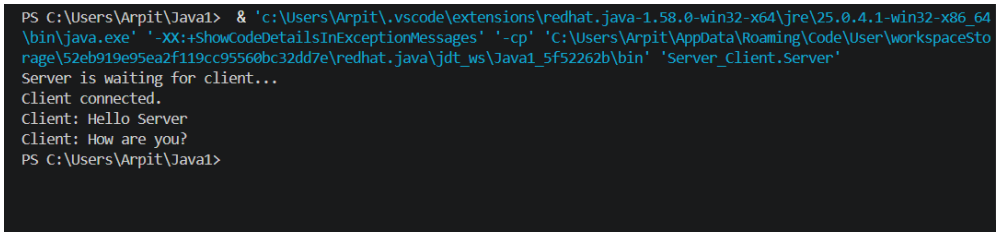
            if (message.equalsIgnoreCase("exit")) {
               break;
            }

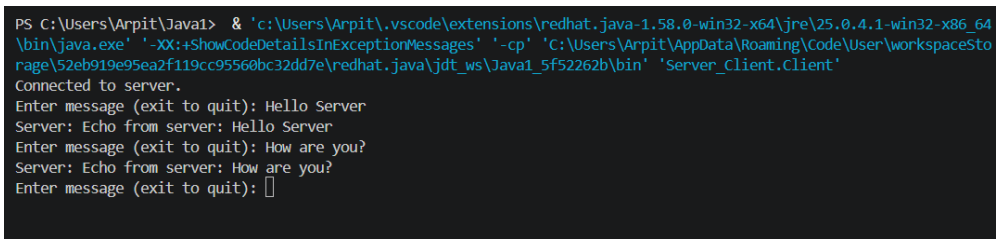
            String response = in.readLine();
            System.out.println("Server: " + response);
         }
```

```
            socket.close();
            sc.close();

        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output Screenshots (if applicable)





Conclusion

Thus, a client-server communication application was successfully implemented using Java socket programming. The client sends messages to the server through a socket, and the server returns the received messages to the client. This practical demonstrates network communication using Socket, ServerSocket, input streams, and output streams in Java.